

# Maxwell T. Elling

PhD Candidate, Atmospheric and Oceanic Sciences  
University of Colorado Boulder  
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## Education

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**PhD in Atmospheric and Oceanic Sciences** 2024–2027 (expected)  
*University of Colorado Boulder*

**MA in Climate and Society** 2020–2021  
*Columbia University*

**BA in Financial Economics** 2015–2019  
*Columbia University*

## Research Experience

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**Graduate Research Assistant** 2024–present  
*University of Colorado Boulder*

**Scientific Programmer / Analyst** 2023–2024  
*NASA Goddard Institute for Space Studies*

**Scientific Programmer** 2021–2023  
*NASA Goddard Institute for Space Studies*

**Graduate Research Assistant** 2021  
*International Research Institute for Climate and Society*

**Graduate Research Assistant** 2021  
*Sabin Center for Climate Change Law*

**Graduate Research Assistant** 2020–2021  
*Lamont Doherty Earth Observatory*

## Peer-Reviewed Publications

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*Published*

9. **Elling, M.T.**, Karneckas, K., Kowalczyk, M. et al.: Tropical oceans drive variability in soil moisture and malaria in Malawi. *Communications Medicine* (2026). <https://doi.org/10.1038/s43856-026-01681-9>
8. **Elling, M.T.**, Ruane, A.C., De Mel, M., Jonas, J., Kadam, S., Kiang, N.Y., Mezuman, K., Nazarenko, L., Pelaccio, N., Phillips, M., Romanou, A., and Rind, D.: An Impact-Driven Framework for Climate Model Evaluation. *Climatic Change* (2026). <https://doi.org/10.1007/s10584-026-04157-w>

7. Amooli, J.A., Miller, R.L., Tsigaridis, K., Chowdhury, S., Zhang, Y., Toolan, C.A., Ahmadi, S., Allen, R.J., **Elling, M.T.**, Ekman, A.M.L., Fraser-Leach, L., Griffiths, P.T., Keeble, J., Koshiro, T., Kushner, P.J., Lewinschal, A., Lund, M.T., MacRae, M., Makkonen, R., Merikanto, J., Nazarenko, L.S., Nabat, P., O'Donnell, D., Oshima, N., Persad, G., Rumbold, S.T., von Salzen, K., Samset, B.H., Swart, N.C., Takemura, T., Wilcox, L.J., and Westervelt, D.M.: Multi-Model Impacts of Dust on African Air Quality and Mortality under Regional and Global Anthropogenic Aerosol Changes. *Journal of Geophysical Research: Atmospheres* (2026). <https://doi.org/10.1029/2025JD046135>
6. Allen, R.J., Carson, T., Liu, W., Wilcox, L.J., Samset, B.H., Ahmadi, S., Ekman, A.M.L., **Elling, M.T.**, Fraser-Leach, L., Griffiths, P., Keeble, J., Koshiro, T., Kushner, P., Lewinschal, A., MacRae, M., Makkonen, R., Merikanto, J., Nabat, P., Nazarenko, L.S., O'Donnell, D., Oshima, N., Paynter, D.J., Persad, G., Rumbold, S., Swart, N., Takemura, T., Tsigaridis, K., von Salzen, K., Westervelt, D.M., and Hassan, T.: AMOC weakening in response to global and regional reductions in aerosol emissions. *Environmental Research: Climate* (2026). <https://doi.org/10.1088/2752-5295/ae63ef>
5. Allen, R.J., Wilcox, L.J., Samset, B.H., Ahmadi, S., Ekman, A.M.L., **Elling, M.T.**, Fraser-Leach, L., Griffiths, P., Keeble, J., Koshiro, T., Kushner, P., Lewinschal, A., MacRae, M., Makkonen, R., Merikanto, J., Nabat, P., Nazarenko, L.S., O'Donnell, D., Oshima, N., Paynter, D.J., Persad, G., Rumbold, S., Swart, N., Takemura, T., Tsigaridis, K., von Salzen, K., and Westervelt, D.M.: Decomposing the global and regional aerosol effective radiative forcing associated with strong versus weak air quality policies by Mid-21st Century. *Environmental Research: Climate* (2026). <https://doi.org/10.1088/2752-5295/ae5418>
4. Kramer, S.M., Karnerauskas, K.B., **Elling, M.T.**, Zhang, L., Liu, H., Chen, Y., Amaya, D.J. et al.: ColdBlobMIP: A Multi-Model Assessment of the Atmospheric Response to the North Atlantic Warming Hole. *Geophysical Research Letters* (2025). <https://doi.org/10.1029/2025GL117784>
3. Nazarenko, L.S., Tausnev, N.L., and **Elling, M.T.**: Temperature and Precipitation Extremes Under SSP Emission Scenarios with GISS-E2.1 Model. *Atmosphere* (2025). <https://doi.org/10.3390/atmos16080920>
2. Yao, Y., Ducharne, A., Cook, B.I., De Hertog, S.J., Aas, K.S., Arboleda-Obando, P.F., Buzan, J., Colin, J., Costantini, M., Decharme, B., Lawrence, D.M., Lawrence, P., Leung, L.R., Lo, M.-H., Devaraju, N., Wieder, W.R., Wu, R.-J., Zhou, T., Jägermeyr, J., McDermid, S., Pokhrel, Y., **Elling, M.**, Hanasaki, N., Muñoz, P., Nazarenko, L.S., Otta, K., Satoh, Y., Yokohata, T., Jin, L., Wang, X., Mishra, V., Ghosh, S., and Thiery, W.: Impacts of irrigation expansion on moist-heat stress based on IRRMIP results. *Nature Communications*, (2025). <https://doi.org/10.1038/s41467-025-56356-1>
1. Samset, B.H., Wilcox, L.J., Allen, R.J., Stjern, C.W., Lund, M.T., Ahmadi, S., Ekman, A., **Elling, M.T.**, Fraser-Leach, L., Griffiths, P., Keeble, J., Koshiro, T., Kushner, P., Lewinschal, A., Makkonen, R., Merikanto, J., Nabat, P., Nazarenko, L., O'Donnell, D., Oshima, N. et al.: East Asian aerosol cleanup has likely contributed to the recent acceleration in global warming. *Communications Earth & Environment* (2025). <https://doi.org/10.1038/s43247-025-02527-3>

#### *In Preparation*

**Elling, M.T.** and Karnerauskas, K.B.: GCHI: A Global Climate Health Index Framework for Quantifying Climate Risks to Human Health. In prep.

**Elling, M.T.** and Karnerauskas, K.B.: Evaluation of Future Global Climate Risks to Human Health. In prep.

**Elling, M.T.** and Karnauskas, K.B.: Evaluation of Future Climate Risks to Human Health in Africa. In prep.

## Conference Papers

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Crompt, S., Rabiei, B., **Elling, M.T.**, Herron, A., Hendrickson, M.: Climate PAL: Climate Analysis through Conversational AI

## Technical Reports

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Carr, E., McCabe, S., **Elling, M.T.**, and Martínez-Loustalot, P.: Emissions and Underwater Noise from LNG Carrier Vessels — A Study of the Potential Impacts of LNG Development on Marine Mammals in the Gulf of California.

Carr, E., McCabe, S., and **Elling, M.T.**: Well-to-Tank Carbon Intensity of Canadian LNG.

Carr, E., McCabe, S., and **Elling, M.T.**: The Impact of Ship Emission Fees on Mode Shift Potential in the United States.

Carr, E., Winebrake, J.J., McCabe, S., and **Elling, M.T.**: Well-to-Tank Carbon Intensity of European LNG Imports.

Carr, E., Winebrake, J.J., McCabe, S., **Elling, M.T.**, and Green, E.H.: Ocean-Going Vessel Decarbonization Technology Assessment.

Carr, E., Winebrake, J.J., McCabe, S., and **Elling, M.T.**: LNG as a Marine Fuel in the United States.

Carr, E., Winebrake, J.J., McCabe, S., and **Elling, M.T.**: LNG and Shipping in the Arctic.

Comer, B., McCabe, S., Carr, E., **Elling, M.T.**, Sturup, E., Knudsen, B., Beecken, J., and Winebrake, J.J.: Real-world NO<sub>x</sub> emissions from ships and implications for future regulations.

Carr, E., McCabe, S., **Elling, M.T.**, and Winebrake, J.J.: Options for Reducing Methane Emissions from New and Existing LNG-Fueled Ships.

Carr, E., Winebrake, J.J., McCabe, S., and **Elling, M.T.**: Black Carbon Emissions from Ships in the Arctic 2019–2024.

## Presentations (First Author & Presenter)

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Elling, M.T., Karnauskas, K.B., Kowalczyk, M. et al., 2026: Tropical oceans drive variability in soil moisture and malaria in Malawi. European Geophysical Union, Annual Meeting 2026. (*Oral*)

Elling, M.T., Karnauskas, K.B., Kowalczyk, M. et al., 2025: Tropical oceans drive Malawi’s malaria risk. American Geophysical Union, Annual Meeting 2025. (*Poster*)

Elling, M.T., Karnauskas, K.B., Kowalczyk, M. et al., 2025: Tropical oceans drive Malawi’s malaria risk. Annual Earth System and Space Science Poster Conference 2025. (*Poster*)

Elling, M.T., Ruane, A.C., Rind, D. et al., 2024: Enhancing Earth System Model Development Through Climate Impact-Driven Model Evaluation. American Geophysical Union, Annual Meeting 2024. (*Poster*)

Elling, M.T., Ruane, A.C., Rind, D. et al., 2023: A Systematic Framework for Investigating Climate Impact-Drivers in NASA GISS ESM Configurations: Towards Improving ESM Development for Impact Applications. American Geophysical Union, Annual Meeting 2023. (*Poster*)

## Peer Review

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*Climatic Change; Environmental Research: Climate; Environmental Research: Health; Geophysical Research Letters*

## Teaching Experience

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**Teaching Assistant**, Polar Ice and Climate University of Colorado Boulder (Fall 2026)

Lectured on Snowball Earth and idealized modeling approaches for evaluating Snowball Earth conditions; created coursework and led laboratory sessions; co-developed experimental materials for measuring snow albedo and physical snow properties; guided students through hands-on experiments

**Guest Lecturer**, Environmental Policy University of Colorado Boulder (Fall 2026)

Lectured on the intersections of climate science, policy, and society, and opportunities for strengthening connections across these domains

## Mentoring

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Zoë Pekarek, undergraduate researcher, University of Colorado Boulder (co-mentor; 2025)

Behrad Rabiei, intern, NASA GISS (co-manager; 2023)

Sonia Crompton, intern, NASA GISS (co-manager; 2023)

Shreya Chandra, intern, NASA GISS (manager; 2022)

## Service

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Co-founder, [Climate Action Knowledge Exchange](#) (2025–present)

Member, Atmospheric and Oceanic Sciences Curriculum Committee (2025/2026)

Member, Atmospheric and Oceanic Sciences Industry Night Committee (2025/2026)

Reviewer, Undergraduate Research Opportunities Program (UROP) proposals (2026)

Organizer, [New Era Network for Societally-Integrated Climatology](#) (2022–2023)

Participant, U.S.–Korea Young Climate Activists Exchange Program (2022)

## Awards

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Atmospheric and Oceanic Sciences Outstanding Service Award (2025/2026)

Annual Earth System and Space Science Poster Conference: 3rd Place (2025)

## Technical Skills

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UNIX/Linux and shell scripting

High-performance computing

Version control (Git)

PostgreSQL and Relational Database Management

Earth System Models: NASA GISS ModelE (code development, setting up and running experiments, CMORization/ESGF publishing/quality control); multi-model CMIP6 suite analysis (dozens of international models)

Open Source Software development: GCHI: A Global Climate Health Index

Programming languages: Python, Fortran, JavaScript, MATLAB, R

## Professional Affiliations

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American Geophysical Union (2022–present)